

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		Femur	1			1		
		(ii)		M=66 or 67/150 (1) = 0.44 or 0.4 / 0.45 or 0.5 (1)		2		2	2	
		(iii)		hydroxyapatite / calcium ions and phosphate ions / calcium phosphate (1)	1			1		
		(iv)		Any two × (1) from: {Haversian/Volkmann} canals contain blood vessels (1) - Diffusion of (named substance/O₂/glucose/amino acids) to {osteoblasts/osteoclasts/ cells} (1) - Canaliculi contains {fluid for exchange with the cells/ tissue fluid} (1)	2			2		
		(v)		Slow fibres are {red /darker} and fast fibres are {white/lighter} (1) {Increased blood supply to/ more blood in} slow twitch fibres / Increased myoglobin in slow twitch fibres/ ORA for fast twitch fibres (1)			2	2		
		(vi)		Total number of fibres in each sample {varied/different} (1) Allows comparison (between different athletes) (1)		2		2		2
		(vii)		the activity type is the only variable that changes/ owtte		1		1		1
		(viii)		{High/ 70} % fast twitch (1) Anaerobic respiration (1) Because there are short intense bursts of activity (1)		1	2	3		
	(b)	(i)		Flexibility / support / protection of spinal cord/ attachment of ribs Any 2 for 1 mark	1			1		
		(ii)		Thoracic		1		1		
		(iii)		Physiotherapy / weight loss / Exercise to strengthen muscles/ or example		1		1		

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		Ball and socket joint	1			1		
		(ii)		Osteoarthritis (1) reduced joint space / reduced cartilage / inflammation /swelling (1)		1	1	2		
				Question 9 total	6	9	5	20	2	2